

Complete Math Roadmap

Canadian Curriculum + ML/CS Math — Restructured

Structure: Learn all math during summers. Do all coding learning during school years.

Golden Rules

- 1. School carries the curriculum. Use self-study time for CS math school skips.
 - 2. Linear algebra before calculus — no calculus needed to start it.
 - 3. Simulate, don't just calculate. Verify every stats/probability concept in Python the same day.
 - 4. The chain rule is backprop. Spend more time here than anything else in calculus.
 - 5. MML book is your capstone math text (free at mml-book.github.io). Read it in Gr. 12 once school has given you the calculus foundation.
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☀ Summer Before Grade 11

~70 days · 1 hr/day math

Math focus: Full MCR3U (Grade 11 Functions) + Linear Algebra start.

Block 1 — MCR3U: Functions (Days 1-30)

Days	Topic	Resource
1-3	Functions, domain/range, notation	Khan Academy — Functions
4-6	Transformations (shift, reflect, stretch)	Khan Academy — Transformations
7-8	Inverse functions	Khan Academy — Inverse Functions
9-11	Quadratics: vertex form, completing the square	Khan Academy — Quadratics
12-13	Quadratic formula, discriminant	Khan Academy — Quadratics
14-15	Polynomial behaviour, factor/remainder theorem	Khan Academy — Polynomials
16-17	Rational functions, asymptotes	Khan Academy — Rational Functions

Days	Topic	Resource
18-21	Exponential functions — growth/decay, transformations	Khan Academy — Exponential Functions
22-23	Solving exponential equations	Khan Academy — Exponential Equations
24-26	Trig — unit circle, radians, CAST rule	Khan Academy — Trigonometry
27-28	Graphing sin/cos/tan, transformations	Khan Academy — Trig Graphs
29-30	Sine law, cosine law	Khan Academy — Trigonometry

Block 2 — Sequences & Series (Days 31-40)

Days	Topic	Resource
31-33	Arithmetic sequences & series	Khan Academy — Sequences
34-36	Geometric sequences & series	Khan Academy — Geometric Sequences
37-38	Infinite geometric series	Khan Academy — Series
39-40	Pascal's triangle, intro to combinatorics	Khan Academy — Combinatorics

Block 3 — Linear Algebra Start (Days 41-65)

Days	Topic	Resource
41-43	Vectors — addition, scalar mult., dot product	3Blue1Brown "Essence of Linear Algebra" Ep. 1-2
44-46	Linear transformations, matrix multiplication	3B1B Ep. 3-4
47-49	Determinants	3B1B Ep. 5-7
50-52	Systems of equations, Gaussian elimination	Khan Academy Linear Algebra
53-55	Matrix inverse, rank, null space	3B1B + Khan Academy

Days	Topic	Resource
56-58	Eigenvectors & eigenvalues	3B1B Ep. 14
59-61	PCA intuition (eigenvectors applied)	StatQuest "PCA" video
62-64	Matrix norms, orthogonality, projections	Khan Academy
65	Review & consolidation	—

Block 4 — Review & Buffer (Days 66-70)

Catch-up days. Re-solve anything shaky. No new material.

Grade 11 School Year

Coding focus: CS50P, C++, DSA habit. Math: already done in summer — school is consolidation only.

School — MCR3U (JensenMath)

You already front-loaded all of this in the summer. School is consolidation and exam accuracy, not new learning.

Unit	Topic	Resource
Unit 1	Functions	jensenmath.ca → Grade 11 → Unit 1
Unit 2	Rational Expressions	jensenmath.ca → Grade 11 → Unit 2
Unit 3	Transformations	jensenmath.ca → Grade 11 → Unit 3
Unit 4	Exponential Functions	jensenmath.ca → Grade 11 → Unit 4
Unit 5	Trig Geometry	jensenmath.ca → Grade 11 → Unit 5
Unit 6	Trig Functions	jensenmath.ca → Grade 11 → Unit 6
Unit 7	Discrete Functions	jensenmath.ca → Grade 11 → Unit 7
Exam	Full Course Review	jensenmath.ca → Grade 11 → Full Course Review

Evenings — Probability & Statistics

~3 sessions/week, 30-45 min. Roughly October → March.

Week	Topic	Resource
1-2	Counting: permutations, combinations	Khan Academy Stats
3-4	Basic probability: sample spaces, events, complement rule	StatQuest + Khan Academy
5-6	Conditional probability, independence	StatQuest
7-8	Bayes' theorem	StatQuest "Bayes' theorem"
9-10	Discrete distributions: Bernoulli, Binomial, Poisson	StatQuest
11-12	Continuous distributions: Uniform, Normal (Gaussian)	StatQuest
13-14	Expectation, variance, standard deviation	Khan Academy
15-16	Central Limit Theorem	StatQuest
17-18	Covariance, correlation	StatQuest
19-20	Maximum Likelihood Estimation (MLE)	StatQuest
21-22	Hypothesis testing, p-values (conceptual)	StatQuest
23-24	Information theory basics: entropy, cross-entropy, KL divergence	Colah's blog

Summer Before Grade 12

~70 days · 1 hr/day math

Math focus: Full MHF4U (Advanced Functions) + MCV4U (Calculus & Vectors) preview + Linear Algebra completion.

Block 1 — MHF4U: Advanced Functions (Days 1-25)

Days	Topic	Resource
1-3	Polynomial functions — graphs, end behaviour, multiplicity of roots	Khan Academy — Polynomials

Days	Topic	Resource
4-6	Dividing polynomials, factor theorem review	Khan Academy — Polynomials
7-9	Rational functions — intercepts, asymptotes (vertical, horizontal, oblique)	Khan Academy — Rational Functions
10-12	Logarithms — definition, laws, change of base	Khan Academy — Logarithms
13-15	Solving log equations, graphing log functions	Khan Academy — Logarithms
16-18	Exponential & log relationship (inverses)	Khan Academy
19-21	Trig identities — Pythagorean, sum/difference formulas	Khan Academy — Trig Identities
22-24	Solving trig equations, general solutions	Khan Academy — Trig Equations
25	Review	—

Block 2 — Calculus Entry Point / MCV4U Preview (Days 26-55)

The chain rule IS backpropagation — go slow, master it.

Days	Topic	Resource
26-28	Limits — intuition, limit laws	Khan Academy — Limits
29-32	Derivatives from first principles	Khan Academy — Derivatives
33-35	Differentiation rules: power, product, quotient	Khan Academy — Differentiation Rules
36-39	Chain rule	Khan Academy — Chain Rule
40-42	Derivatives of exponentials & logs (e^x , $\ln x$)	Khan Academy
43-45	Derivatives of trig functions	Khan Academy
46-48	Partial derivatives — intro	Khan Academy — Multivariable
49-51	Gradients & gradient vectors	Khan Academy — Multivariable
52-54	Optimization — critical points, 2nd derivative test	Khan Academy — Optimization
55	Review	—

Block 3 — Linear Algebra Completion + Multivariable (Days 56-70)

Days	Topic	Resource
56-58	Jacobian matrix intuition	MML Book Ch. 5
59-61	Gradient descent implemented in NumPy	—
62-64	SVD intuition (Singular Value Decomposition)	StatQuest "SVD"
65-67	Vector calculus: gradient, divergence (light)	MML Book
68-70	Backpropagation from scratch — derive it mathematically	Karpathy "Micrograd"

Grade 12 School Year

Coding focus: DevOps, System Design, Classical ML. Math: already done in summer — school is consolidation only.

School — MHF4U Advanced Functions (JensenMath)

You've already previewed this entire course. School is consolidation and exam accuracy.

Unit	Topic	Resource
Unit 1	Polynomial Functions	jensenmath.ca → Grade 12 → MHF4U → Unit 1
Unit 2	Rational Functions	jensenmath.ca → Grade 12 → MHF4U → Unit 2
Unit 3	Exponential & Logarithmic Functions	jensenmath.ca → Grade 12 → MHF4U → Unit 3
Unit 4	Trigonometric Functions & Identities	jensenmath.ca → Grade 12 → MHF4U → Unit 4
Unit 5	Rates of Change	jensenmath.ca → Grade 12 → MHF4U → Unit 5
Exam	Full Course Review	jensenmath.ca → Grade 12 → MHF4U → Full Course Review

School — MCV4U Calculus & Vectors (JensenMath)

You've previewed derivatives and the chain rule. School fills in the full picture.

Unit	Topic	Resource
Unit 1	Derivative Rules	jensenmath.ca → Grade 12 → MCV4U → Unit 1
Unit 2	Curve Sketching	jensenmath.ca → Grade 12 → MCV4U → Unit 2
Unit 3	Derivative of Trig & Exponentials	jensenmath.ca → Grade 12 → MCV4U → Unit 3
Unit 4	Geometric Vectors	jensenmath.ca → Grade 12 → MCV4U → Unit 4
Unit 5	Algebraic Vectors	jensenmath.ca → Grade 12 → MCV4U → Unit 5
Unit 6	Lines & Planes	jensenmath.ca → Grade 12 → MCV4U → Unit 6
Exam	Full Course Review	jensenmath.ca → Grade 12 → MCV4U → Full Course Review

Evenings — ML Math Consolidation

~3 sessions/week, 30–45 min. Mathematics for Machine Learning (free at mml-book.github.io) is the capstone text.

Period	Topic	Resource
Sept-Oct	Multivariate calculus — chain rule in matrix form, Jacobians	MML Book Ch. 5
Nov-Dec	Optimization theory — convexity, saddle points, Adam/momentum math	MML Book Ch. 7
Jan-Feb	Linear algebra deep dive — SVD, matrix factorization, spectral methods	MML Book Ch. 2-4
Mar-Apr	Probability theory formal — measure theory lite, conjugate priors, distributions in depth	MML Book Ch. 6
May-Jun	Information theory — entropy, mutual information, KL divergence applications	Elements of Information Theory Ch. 1-2

☀ Summer After Grade 12

~70 days — aligns with Phases 7-8: Deep Learning + Applied AI.

This is where the math pays off: implementing backprop, attention mechanisms, and loss functions for real.

Block 1 — Deep Learning Math (Days 1-30)

Days	Topic	Resource
1-5	Backprop derivation — scalar $\rightarrow$ vector $\rightarrow$ matrix form	Karpathy "Micrograd" + MML Ch. 5
6-10	Loss functions derived — MSE, cross-entropy, KL divergence	MML Book
11-15	Activation functions — sigmoid, tanh, ReLU, derivatives	MML Book + d2l.ai
16-20	Attention mechanism math — scaled dot-product, softmax derivation	3B1B "Attention" + d2l.ai
21-25	Convolutional math — cross-correlation, filter math	d2l.ai CNN chapter
26-30	Regularization math — L1/L2 as priors (Bayesian view), dropout	MML Book Ch. 8

Block 2 — Advanced CS Math (Days 31-55)

Days	Topic	Resource
31-35	Combinatorics & counting — generating functions, recurrences	Competitive Programmer's Handbook
36-40	Graph theory — trees, cycles, planarity, graph representations	USACO Guide Gold + CP Handbook
41-45	Discrete math — logic, sets, relations, proofs	CP Handbook
46-50	Number theory — modular arithmetic, Fermat's little theorem, RSA	CP Handbook — number theory chapters
51-55	Big-O formal — asymptotic notation proofs, recurrence relations (Master theorem)	CP Handbook + CLRS reference

Block 3 — Bayesian & Statistical Learning Theory (Days 56–70)

Days	Topic	Resource
56–59	Bayesian ML — Bayesian linear regression, MAP estimation	MML Book Ch. 9
60–63	Statistical learning theory — bias-variance tradeoff formally, VC dimension (intro)	MML Book + ESL reference
64–67	Embedding math — vector spaces, cosine similarity derivation, why dot product works	MML Book
68–70	Review + fill gaps from Phase 7–8 coding work	—

Complete Curriculum Checklist

Ontario School Curriculum

- ☐ MPM1D/2D (Gr. 9–10) — assumed done
- ☐ MCR3U (Gr. 11 Functions) — Summer before Gr. 11 + school year consolidation
- ☐ MHF4U (Gr. 12 Advanced Functions) — Summer before Gr. 12 + school year consolidation
- ☐ MCV4U (Gr. 12 Calculus & Vectors) — Summer before Gr. 12 + school year consolidation
- ☐ MDM4U (Data Management — optional) — covered by stats self-study, better than the course

ML/CS Math

- ☐ Linear Algebra — Summer Gr. 11, finished Gr. 12 evenings
- ☐ Probability & Statistics — Gr. 11 evenings
- ☐ Calculus (derivatives, chain rule, partial derivatives) — Summer Gr. 12
- ☐ Multivariable Calculus / Gradient Descent math — Summer Gr. 12
- ☐ Information Theory — Gr. 12 evenings
- ☐ Backprop math — Summer after Gr. 12
- ☐ Attention math — Summer after Gr. 12
- ☐ Discrete Math / Combinatorics — Summer after Gr. 12

☐ Graph Theory — Summer after Gr. 12

☐ Number Theory — Summer after Gr. 12